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Root Cause Analysis report
Nghi Son2 BFP Motor

2026.02.19

HYOSUNG HEAVY INDUSTRIES INDUSTRIAL MACHINERY PU
CUSTOMER SERVICE TEAM

Contents

Ⅰ. Executive Summary

Ⅱ. Inspection

Ⅲ Review of Root Cause Analysis

Ⅳ. Conclusion

APPENDIX

Ⅰ. TEST REPORT
Ⅱ. Improvement
Ⅲ. Repair Schedule

Ⅰ. Executive Summary

1. Summary

A. Customer : Nghi Son2, Vietnam, DOOSAN

B. Application : Unit #2, Boiler feedwater pump 2B

C. Specification : 10111700_0004 / 14,600kW_4P_11,000V_50Hz_876[A]

D. Quantity supplied : Total 5 units (1 unit failure, 4 units in operation)

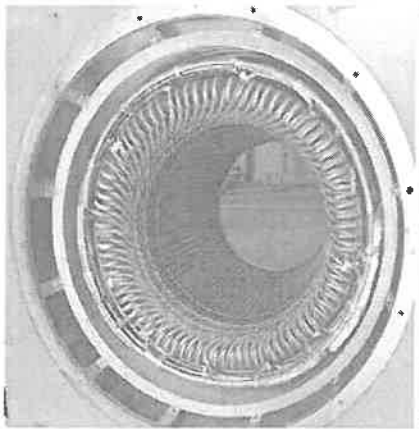
2. History

Date	Event / Observation
02-Nov-2025	- During normal operation at a minimum load of 267 MW, BFP-2B motor tripped due to differential protection (87) operation. - Insulation resistance (IR) measurement of motor windings (phase-to-phase and phase-to-ground) was less than 1 MΩ. - Inspection confirmed blackening and carbonization at the DE-side winding end, arc marks, soot deposits, and insulation deterioration.
05-Nov-2025	The failed motor was replaced with a spare motor.
23-Jan-2026	The motor arrived at the repair shop.
02-Feb-2026	A joint disassembly inspection was conducted by KEPRI, DOOSAN, and HYOSUNG.

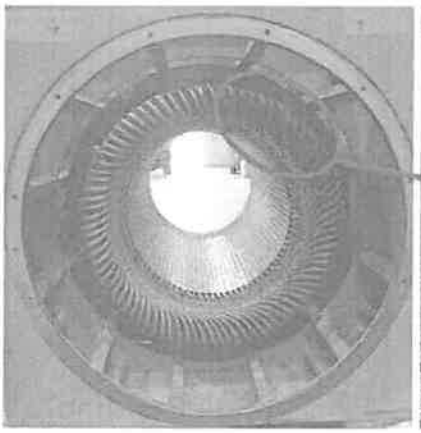
Ⅱ. Inspection

1. Visual Inspection

- The overall internal condition of the stator windings is good.
- The non-drive end (NDE) windings are in good condition, with no contamination from carbonized dust.
- The drive end (DE) windings are contaminated with carbonized dust, and black contamination was particularly observed in the 12 to 3 o'clock direction section.



(Fig 1) View from NDE side



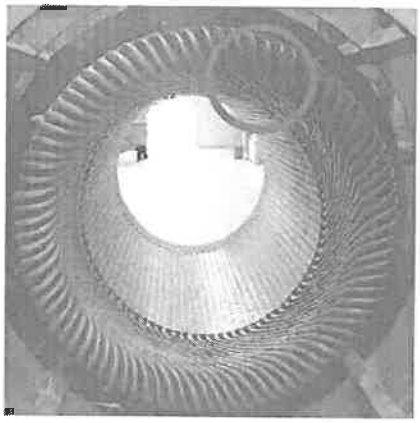
(Fig 2) View from DE side

(Fig 3) 12~3 o'clock area

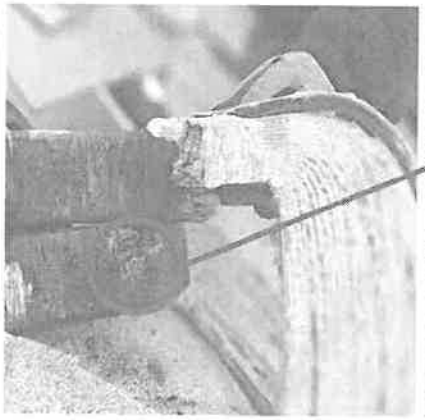
Ⅱ. Inspection

1. Visual Inspection

- The fault location was identified at the lower side of the first upper coil of phase A on the drive end (DE).
- For a detailed inspection, the winding was cut, revealing damage to the upper coil just below the wedge.
- Based on the nature of the damage, it is judged that a turn-to-turn short occurred.



(Fig 4) Failure Location



(Fig 5) Failure Location Detail

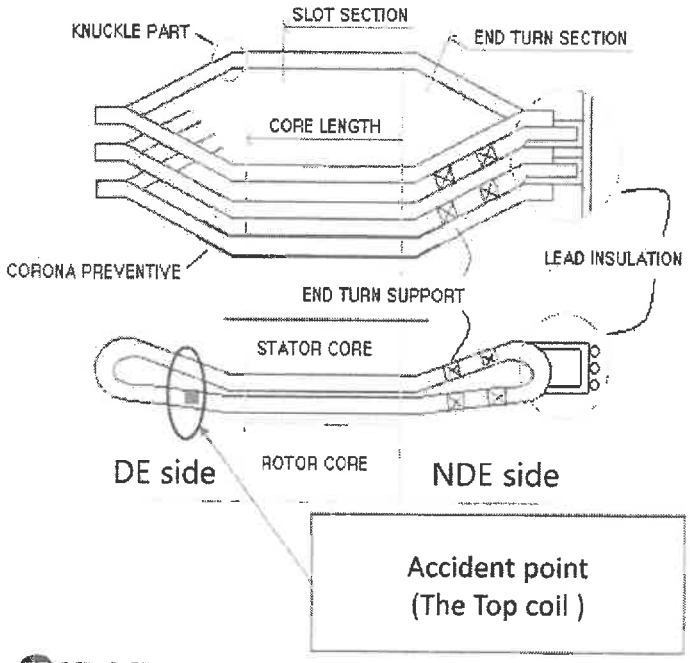
(Fig 6) Failure Condition After Main Insulation Removal

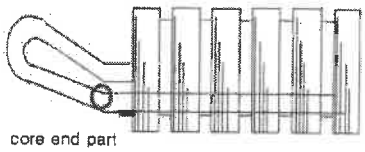
Ⅱ. Inspection

2. Coil cross section of damaged part

A. Opinion for inspection on damaged coil part

- The incident was confirmed to have occurred at the lower side of the top coil of the drive end (DE) winding end.





(Fig 7) Coil cross section

Ⅱ. Inspection

2. Coil cross section of damaged part

B. Opinion for inspection on damaged coil part

- The incident was confirmed to have occurred at the lower side of the top coil of the drive end (DE) winding end.

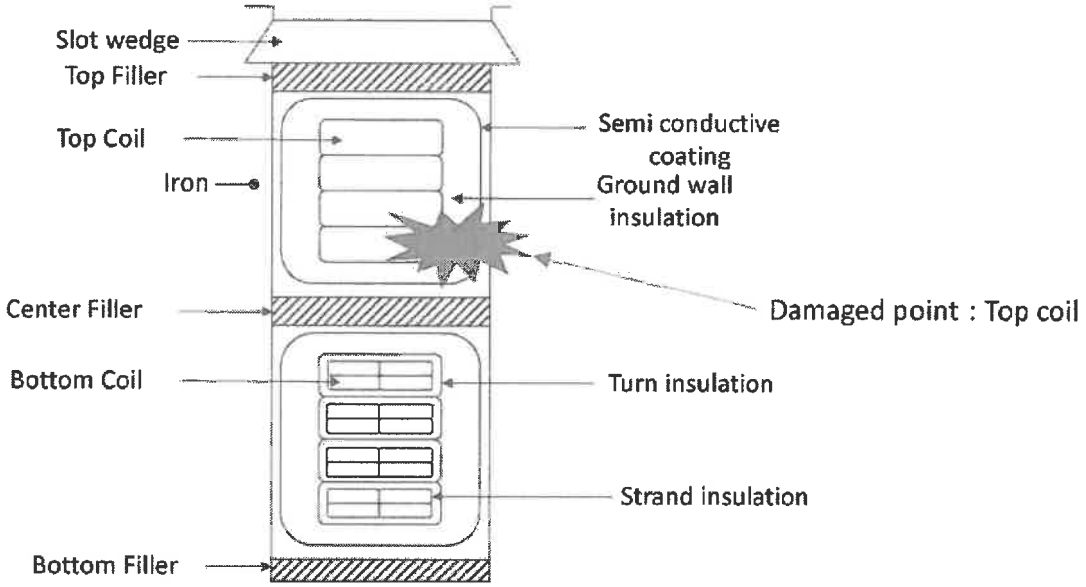


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Fig 8. Coil cross section

Ⅲ Review of Root Cause Analysis

1. Design & Operation condition : The design review has been completed in accordance with the design standards, and it is presumed that the damage was influenced by surge voltage.

Classfication	Cause(I)	Cause(II)	Cause(III)	Possibility	Verification Method(I)
Design	Specification	Fault Level	satisfy 50kA/0.5Sec?	X	Conductor cross-sectional area and lead wire thickness selected according to fault level specifications
	Design review	Electrical review	Are the appropriate insulation standards applied for 11kV rating?	X	Application of 11kV class insulation cord
			Is there any effect on the vibration generated in the core?	X	Confirmation of satisfaction of core amplitude design criteria
Operating condition	Power Quality	Surge Voltage Generation	Check whether surge voltage occurs	Δ	Check the surge event records insulation breakdown pattern of the damaged winding

Ⅲ Review of Root Cause Analysis

2. Manufacturing : No abnormalities were found during the inspection.

Classfication	Cause(I)	Cause(II)	Cause(III)	Possibility	Verification Method(I)
Manufacturing	Material	Conductor Insulation	Compliance with Design Specifications for Turn Insulation	X	Produced in accordance with design specifications; no issues found
		Main Insulation	Compliance with Design Specifications for Main Insulation	X	Produced in accordance with design specifications; no issues found
	Dimensions	Stator	Compliance With Design Drawings for Stator Manufacturing Dimensions	X	Produced in accordance with design specifications; no issues found
		Insulation Clearance	Compliance With Design Specifications for Insulation Clearance	X	Produced in accordance with design specifications; no issues found
	Method	VPI	Compliance With Manufacturing Standards for VPI Process	X	Produced in compliance with manufacturing standards no issues found
	Test	Resistance	Compliance With Phase-to-Phase and Phase-to-Ground Resistance	X	Measured resistance values within acceptable range no issues found
		Insulation Resistance	Compliance With Insulation Resistance	X	Insulation resistance complies with specifications; no issues found
		Dielectric Withstand	Pass/Fail Status of Dielectric Withstand Test	X	Withstand voltage test passed; no issues found
		Insulation Diagnosis	Compliance With Insulation Diagnosis Values	X	Insulation diagnosis values meet criteria no issues found
		Surge Test	Pass/Fail Status of Surge Test	X	Surge test passed; no issues found

Ⅲ Review of Root Cause Analysis

3. Qualification : No abnormalities were found during the inspection.

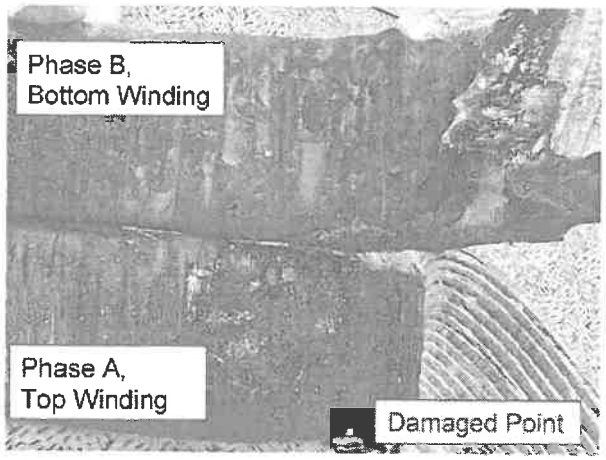
Classfication	Cause(I)	Cause(II)	Cause(III)	Possibility	Verification Method(I)
Qualification	Insulation integrity	Is the insulation Integrity test Method appropriate?	Have you performed a voltage Test ?	X	We performed a voltage test through customer attendance and in-house testing, and found no unusual issues.
			Have you measured the insulation resistance ?	X	After the voltage test, the insulation resistance was measured and the standard Was met
			Have you performed additional Insulation integrity(insulation Diagnosis) checks?	X	Insulation diagnosis(AC test, partial discharge Test, Tan delta test) was performed and the Manufacturer's standards were met.

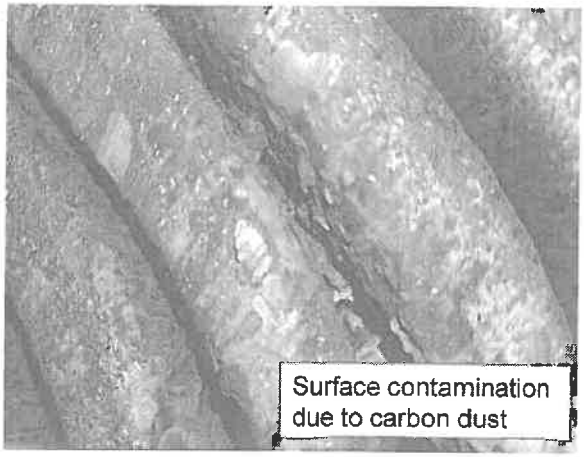
Ⅲ Review of Root Cause Analysis

4. Root Cause of accident

A. Initial Defect Occurrence(Turn-to-turn Short Initiation)

A turn-to-turn short occurred at the corner approximately one-third of the way from the core side along the entire length of the ECP (Endwinding Corona Protection) on the lower side of the first top coil of Phase A (at the contact surface of the lower coil). This resulted in the formation of smoke and carbonized dust, and the surrounding area became blackened and soiled.





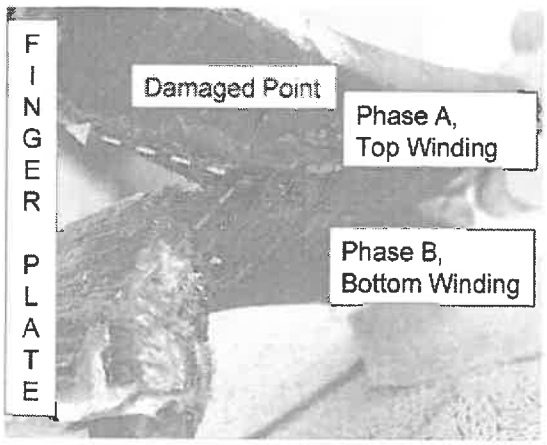
Ⅲ Review of Root Cause Analysis

4. Root Cause of accident

B. Damage Progression Mechanism(Surface Tracking Propagation)

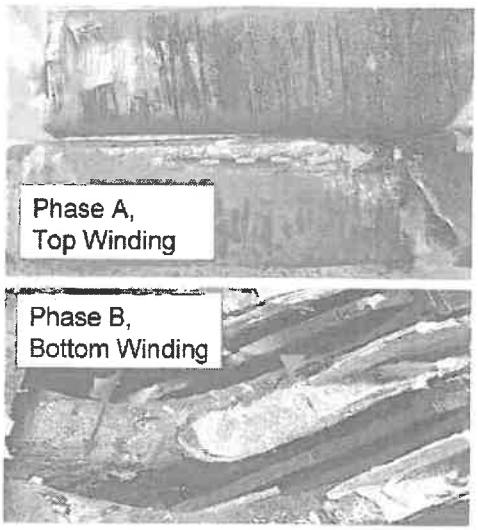
The carbonized dust generated by the turn-to-turn short acted as a conductive material, forming an electrical path for surface tracking and accelerating its propagation. Surface tracking initiated at the location of the turn-to-turn short (power side) and progressed almost simultaneously in two directions as follows:.

- corner of Phase A-> top of the end plate
- contact surface of the lower coil of Phase B-> bottom of the end plate



Tracking path (Phase A)

Tracking path (Phase B)

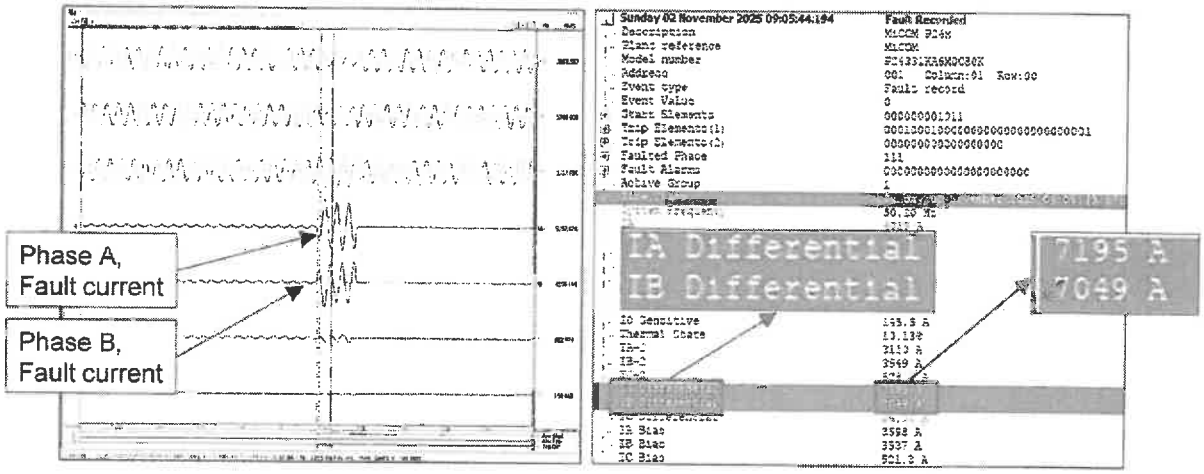


Ⅲ Review of Root Cause Analysis

4. Root Cause of accident

C. Final Failure Result (Ground Fault & Melting)

The surface tracking at the corner of Phase A (upper side) and the surface tracking on the lower side of Phase B occurred almost simultaneously with the end plate, resulting in a ground fault. The ground fault caused a sudden surge of current, which led to melting of the copper conductor and partial exposure of copper to the outside. The fault analysis waveforms also confirmed the sudden current surge in Phases A and B.



Ⅲ Review of Root Cause Analysis

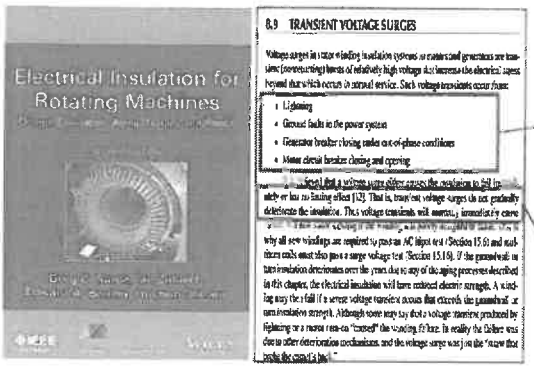
4. Root Cause of accident

D. Presumed Cause

The upper winding where the turn-to-turn short occurred corresponds to the first coil at the incoming end where phase voltage is initially applied. When an external surge enters, turn shorts tend to occur most frequently at the first coil where voltage is first applied. In particular, surge voltages that rise rapidly over a short period are a very hazardous factor for damaging turn insulation in windings with multi-turn coil structures, such as those used in high-voltage motors.

Additionally: **a)** Manufacturing defects in the winding often lead to immediate failure during initial operation; however, since the failure occurred approximately three years into service, the likelihood of a manufacturing defect is considered low. Therefore, it is presumed that turn insulation degradation caused by external surges may have been the initial cause.

b) Major causes of external surges include lightning , ground faults in the power system, generator breaker closing under out-of phase conditions, motor circuit breaker closing and opening.



Reference : Electrical Insulation for Rotating Machines

a)

- Lightning
- Ground faults in the power system
- Generator breaker closing under out-of-phase conditions
- Motor circuit breaker closing and opening

b)

deteriorate the insulation. Thus voltage transients will normally immediately cause failure in a new stator winding if the winding was poorly designed or made. This is

IV. Conclusion

1. Estimated Root Cause

A. No anomalies related to this incident were identified from the perspectives of design, manufacturing, or testing.

B. The failure occurred at the first upper winding where phase voltage is initially applied, which is the most vulnerable location to external surges. It is possible that turn insulation gradually degraded due to transient overvoltages entering during operation.

C. The carbonized dust generated after the initial turn-to-turn short acted as a conductive path, allowing surface tracking to spread in both directions (toward the Phase A corner and the lower coil contact surface of Phase B), which ultimately caused a ground fault with the end plate.

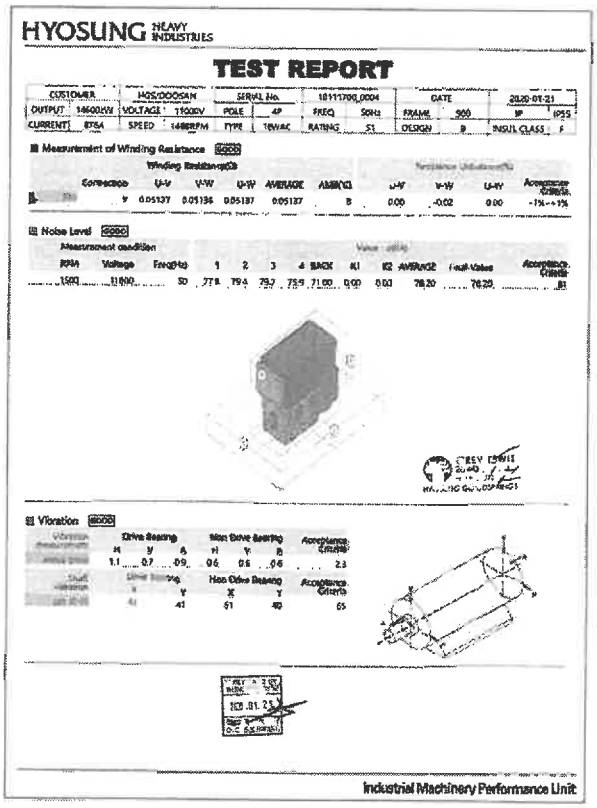
D. The motor operated normally for approximately three years before the failure occurred, which does not correspond with the characteristics of typical manufacturing defects (early failure during initial operation). Therefore, it is considered more likely that the incident was caused by external operational factors (such as lightning, system abnormalities, or breaker operation) rather than a manufacturing defect.

2. Action Plan

A. Conduct insulation diagnostics on the four motors currently in operation to assess the condition of the windings.

APPENDIX I

1. Test Report



HYOSUNG HEAVY INDUSTRIES

TEST REPORT

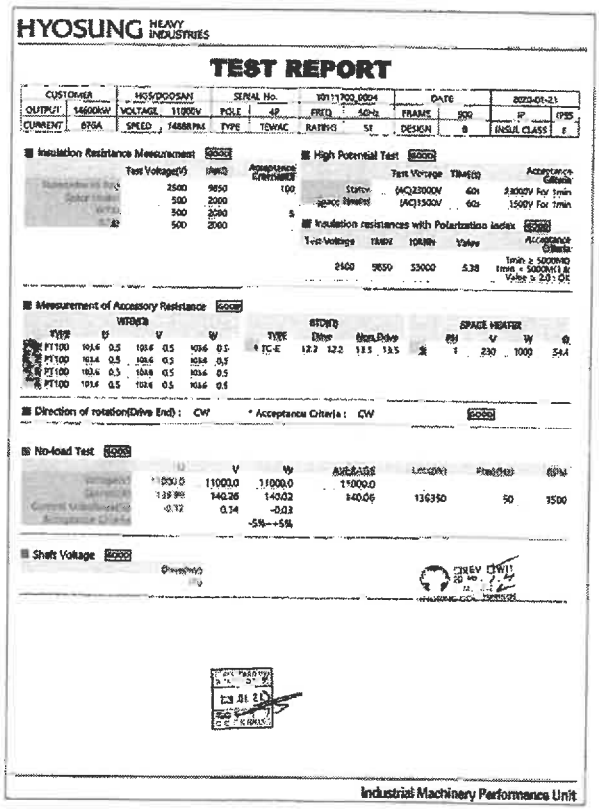
CUSTOMER	HOSYDOOSAN	SERIAL No.	101111700_0004	DATE	2020-01-21
OUTPUT 14600kW	VOLTAGE 11000V	POLE 4P	FREQ 50Hz	FRAME 900	IP IP55
CURRENT 875A	SPEED 1488RPM	TYPE TEWAC	RATING S1	DESIGN B	INSUL CLASS F

Measurement of Winding Resistance [GOOD]

Noise Level [GOOD]

Vibration [GOOD]

Industrial Machinery Performance Unit



HYOSUNG HEAVY INDUSTRIES

TEST REPORT

CUSTOMER	HOSYDOOSAN	SERIAL No.	101111700_0004	DATE	2020-01-21
OUTPUT 14600kW	VOLTAGE 11000V	POLE 4P	FREQ 50Hz	FRAME 900	IP IP55
CURRENT 875A	SPEED 1488RPM	TYPE TEWAC	RATING S1	DESIGN B	INSUL CLASS F

Insulation Resistance Measurement [GOOD]

High Potential Test [GOOD]

Insulation resistances with Polarization Index [GOOD]

Measurement of Accessory Resistance [GOOD]

Direction of rotation(Drive End) : CW * Acceptance Criteria : CW [GOOD]

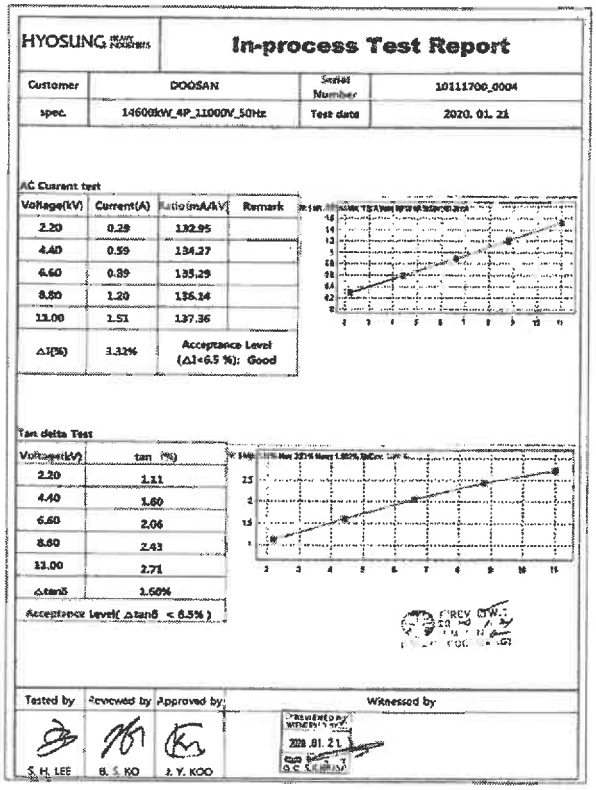
No-load Test [GOOD]

Shaft Voltage [GOOD]

Industrial Machinery Performance Unit

APPENDIX I

1. Test Report



HYOSUNG HEAVY INDUSTRIES — **In-process Test Report**

Customer	DOOSAN	Serial Number	10111700_0004
spec.	14600kW_4P_11000V_50Hz	Test date	2020. 01. 21

AC Current test

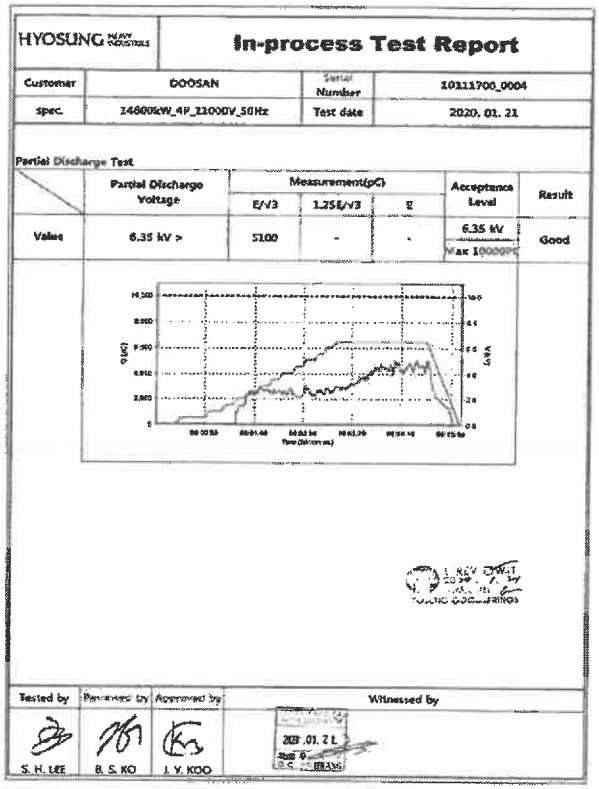
Voltage(kV)	Current(A)	Ratio(mA/kV)	Remark
2.20	0.29	132.95	
4.40	0.59	134.27	
6.60	0.89	135.29	
8.80	1.20	136.14	
11.00	1.51	137.36	
ΔI(%)	3.32%	Acceptance Level (ΔI<6.5 %): Good	

Tan delta Test

Voltage(kV)	tan (%)
2.20	1.11
4.40	1.60
6.60	2.06
8.80	2.43
11.00	2.71
Δtanδ	1.60%

Acceptance level(Δtanδ < 6.5%)

Tested by	Reviewed by	Approved by	Witnessed by
S. H. LEE	B. S. KO	J. Y. KOO	



HYOSUNG HEAVY INDUSTRIES — **In-process Test Report**

Customer	DOOSAN	Serial Number	10111700_0004
spec.	14600kW_4P_11000V_50Hz	Test date	2020. 01. 21

Partial Discharge Test

	Partial Discharge Voltage	Measurement(pC)			Acceptance Level	Result
		E/√3	1.25E/√3	E		
Value	6.35 kV >	5100	-	-	6.35 kV Max 10000pC	Good

Tested by	Reviewed by	Approved by	Witnessed by
S. H. LEE	B. S. KO	J. Y. KOO	

APPENDIX II. Improvement

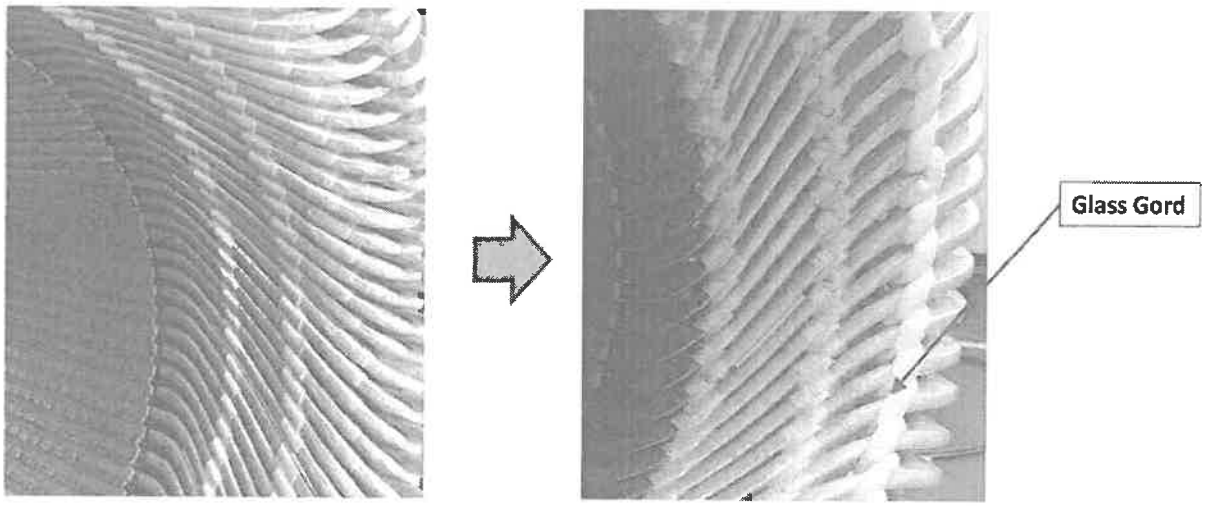
■ Comparison Table of Insulation Materials

Item		Previous Insulation System	Improved Insulation System	Remarks
Insulation Class		13.8 kV_Class F	13.8 kV_Class F	Same
Strand Insulation Mica Tape	Manufacturer	NRK	NRK	A Semi-Dry type mica tape with higher resin content than the previous Dry type is applied. This improves resin penetration during the VPI process and minimizes voids within the insulation layer, thereby reducing the likelihood of partial discharge.
	Type	Dry Type	Semi-Dry Type	
	Thickness	0.065mm	0.06mm	
	Application	1/3 Overlap, 2 Turn	1/3 Overlap, 2 Turn	
Main Insulation Mica Tape	Manufacturer	NRK	NRK	A mica tape incorporating glass chop is applied, forming micro-channels inside the tape. This allows resin to penetrate uniformly into the insulation layer during the VPI process, achieving a void-free and enhancing insulation reliability.
	Thickness	0.18mm	0.135mm	
	Application	1/2 Overlap, 7 Turn	1/2 Overlap, 9 Turn	
OCP	Manufacturer	Vonroll	Vonroll	An upgraded product with improved aging and partial discharge characteristics compared to the previous material
	Thickness	0.085mm	0.09mm	
	Application	1/2 Overlap, 1 Turn	1/2 Overlap, 1 Turn	

APPENDIX Ⅱ. Improvement

■ Improved Coil Support Structure

Glass cord has been additionally applied to the coil end to increase the mechanical support rigidity of the winding.
This enhancement improves the support structure and minimizes displacement due to electromagnetic forces and vibration during operation.

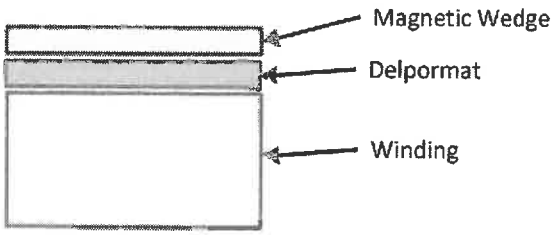


(The image below is for illustrative purposes only and may vary from the actual product)

APPENDIX Ⅱ. Improvement

■ Enhanced Wedge Fixation Strength

1. Modification of Delpormat material specification
   - Delpormat is applied to the upper section of the slot, and its thickness expansion during the VPI impregnation process enhances the bonding and contact pressure within the slot.
   - The Delpormat thickness has been increased to further enhance the contact pressure



2. Air-dry varnish coating
   - After completing the VPI impregnation, additional air-dry varnish is applied to the wedge contact area and the top of the slot to further reinforce the wedge fixation strength

APPENDIX Ⅲ. Repair Schedule

BFP Motor Repair Schedule

2026-02-04

| Step | | February | | | | | | | | | | | | | | | | | | | | | | | | | | | | | March | | | | | | | | | | | | | | | | | | |